

Odonata (Insecta) as a tool for the biomonitoring of environmental quality

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ABSTRACT

Despite the fundamental dependence of human populations on water resources, a range of anthropogenic impacts, in particular the removal of riparian vegetation, threaten freshwater environments. One of the most effective means of evaluating the effects of anthropogenic disturbance in aquatic ecosystems is the use of bioindicators, and the insects of the order Odonata are among the most efficient indicators, due to their enormous sensitivity to environmental changes. In this context, the present study aimed to verify which parameters of the odonate community (species richness, abundance/biomass, composition, taxonomic diversity and taxonomic/phylogenetic distinctness) are most effective for the evaluation of the loss of environmental integrity. The study focused on 50 streams in the northeast of the Brazilian state of Pará. The streams were sampled during the dry season, between June and August 2011. The physical characteristics of each stream were evaluated using a Habitat Integrity Index (HII). The species composition provided the best parameter for the evaluation of ecological integrity, providing a relatively accurate assessment at a lower mean research cost than other parameters. Taxonomic diversity and distinctness also provided relatively reliable results, contributed additional information on the evolutionary relationships among the odonate taxa, and also provided a low-cost approach. Deconstructing communities is necessary to detect impacts, considering the considerable variation in the environmental requirements of the different species. Overall, the parameter that best responded to gradients of disturbance was species composition, followed by diversity and taxonomic distinctness. Given these findings, odonate-based biomonitoring should focus on these parameters to guarantee the optimal detection and evaluation of habitat alterations.

1. Introduction

Despite the fundamental dependence of human populations on aquatic resources, freshwater ecosystems are among the most threatened environments on Earth (Dudgeon et al., 2006; Vörösmarty et al., 2010). The loss of biodiversity in freshwater environments may be up to five times higher than in terrestrial environments (Sala et al., 2000; Dudgeon et al., 2006; Dudgeon, 2010). A number of different types of anthropogenic impact affect freshwater ecosystems, in particular the disturbance of substrates. These processes alter natural environmental gradients (Couceiro et al., 2012) and reduce the variability of habitats by homogenizing conditions (Bleich et al., 2015). This disruption leads to a shift in the natural dynamics of biological communities (Goulart and Callisto, 2003; Couceiro et al., 2011, 2012).

The removal of the riparian vegetation from areas bordering aquatic ecosystems aggravates these impacts, and affects aquatic communities by increasing the input of sediment into the bodies of water (Nakamura

and Yamada, 2005). The absence of marginal vegetation also results in a reduction in the amount of debris and organic matter found in the stream bed (Price et al., 2003), and an increase in the flow of water and the penetration of sunlight (Macdonald et al., 2003). This favors primary production, leading to an increase in the availability of prey for predatory insects, such as odonates (Oliveira Junior et al., 2013; Carvalho et al., 2013).

In aquatic systems, one of the most effective means of evaluating anthropogenic is the use of bioindicators, which are organisms or communities with functional characteristics so closely related to specific environmental factors that any change in the environment can provoke a predictable shift in the biota (Silva et al., 2010). These bioindicators may respond to habitat change on multiple spatial and temporal scales, including land use (Ometo et al., 2000; Sponseller et al., 2001; Ligeiro et al., 2013; Macedo et al., 2014), riparian vegetation (Monteiro-Júnior et al., 2013; Oliveira Junior et al., 2015), habitat complexity (Weigel et al., 2003), and physical and chemical

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variables of the water (Anjos and Takeda, 2005). The insects of the order Odonata are particularly effective bioindicators for the assessment of aquatic environments. These insects occupy the interface between aquatic and terrestrial ecosystems and are highly sensitive to environmental changes (Carvalho et al., 2013; Monteiro-Júnior et al., 2013; Oliveira Junior et al., 2015), and are extremely diverse in tropical regions (Kalkman et al., 2008). There is also a major division within the order in terms of the ecophysiological requirements of the different species (De Marco et al., 2015), and the structure of odonate communities can shift predictably in response to changes in local environmental conditions (Corbet, 1999; Juen et al., 2007; Juen and De Marco, 2011; Monteiro-Júnior et al., 2013). In particular, disturbance of the riparian vegetation has a direct effect on the structure of the adult odonate community, given that ambient temperatures have a direct influence on the occurrence of species and their reproductive cycles (Samways and Steytler, 1996). Reproductive behavior and mate choice is also impaired, given that females select males that guard the best oviposition sites in the riparian vegetation (De Marco and Resende, 2004).

Understanding patterns of species diversity related to the quality of the environment is crucial to the definition of adequate strategies for the conservation of biological diversity (Veech et al., 2002). One widely-used approach to the assessment of ecological integrity is the application of indices of biological integrity (IBI). This approach was originally proposed by Karr (1981) for North American fish communities, but it is now used in many different contexts worldwide. These indices are compiled by comparing the values of biological metrics recorded in highly degraded areas with those obtained under reference conditions, typically the least impacted habitats found within the local region (Hughes, 1995). The IBI approach, and the use of multimetric indices, has become extremely popular in North America (Angermeier and Karr, 1986; Miller et al., 1988; Whittier et al., 2007; Pyron et al., 2008), Europe (Oberdorff and Hughes, 1992; Angermeier and Davideanu, 2004), Central America (Lyons et al., 1995), India (Ganasan and Hughes, 1998), Africa (Kamdem Toham and Teugels, 1999) South America (Moya et al., 2011; Araújo et al., 2003; Bozzetti and Schulz, 2004; Terra et al., 2013; Macedo et al., 2016), and New Zealand (Joy and Death, 2004). This tool has allowed conservationists and policy makers to establish systematic priorities for biodiversity conservation in many regions (Pont et al., 2006; Ruaro and Gubiani, 2013).

The taxonomic richness and relative abundance of species are among the biological variables used most frequently in environmental assessments and multimetric indices (Clarke and Warwick, 2001), although the reliable measurement of species richness is a major problem (Gotelli and Cowell, 2001). Taxonomic richness also attributes equal importance to all species, regardless of their conservation value (Cianciaruso et al., 2009). In fact, taxonomic richness may not be sensitive to environmental impacts at all in communities where the reduction of the richness of the most sensitive species is counterbalanced by an increase in the number of more generalist species, which are resistant to changes in the environment (Carvalho et al., 2013). This is the reason why species richness and abundance are not the only biological variables used to measure environmental integrity.

The “ABC curve” (Abundance Biomass Comparison) is another method of detecting anthropogenic impacts (Warwick, 1986). This approach is based a detailed comparison of the distribution of abundance and biomass among the different species in each sample, based on the classical evolutionary theory of *r* and *K* selection (Clarke and Warwick, 2001). In pristine environments, communities are dominated by *K* strategists, dominated by relatively large organisms, resulting in a biomass curve entirely located above the abundance curve. When the system is disturbed, it will favor the emergence of *r*-strategists, with a predominance of relatively small individuals, and an abundance curve entirely above the biomass curve (Warwick, 1986). While this is a potentially lucrative approach, no study has yet applied it to odonate communities.

Many indices of biodiversity do not include information on the number of species or individuals in the communities (Cianciaruso et al., 2009). Metrics such as taxonomic diversity (Δ) and distinctness (Δ^*), which were initially used to assess the response of benthic communities to anthropogenic disturbance (Warwick and Clarke, 1995, 1998) are highly robust, given that they take into account the phylogenetic relationships of the species based on the taxonomic classification of pairs of species in the community. These metrics are also less dependent on sampling effort because they are based on non-quantitative (presence or absence) data and are easily calculated and applied to the simultaneous analysis of a number of different communities (Clarke and Warwick, 1998; Magurran, 2004).

Given the current scenario of environmental degradation and the lack of reliable data on the distribution of most species, the development of effective strategies for the rapid and accurate assessment of ecosystems is an important priority. The present study investigated and evaluated the different types of univariate and multivariate assemblage characteristics that can be used to assess the loss of environmental integrity, with the primary objective of identifying the parameters that provide the most effective measure for the detection of impacts. The metrics tested here are standard measures used in ecological studies, including taxonomic richness, ABC curves, taxonomic composition, taxonomic diversity, and the taxonomic/phylogenetic distinctness of the target odonate assemblages.

2. Material and methods

2.1. Study area

The study was carried out in 50 stream reaches in the municipality of Paragominas, located in the northeastern portion of the state of Pará, Brazil (2°25′–4°09′ S, 46°25′–48°54′ W) (Watrin and Rocha, 1992) (Fig. 1). The predominant climate of the study region is of the “Aw” type in the Köppen classification system, that is, tropical rainy, with a short, well-defined dry season, with monthly precipitation below 60 mm, mean annual temperature of 27.2 °C, relative humidity of 81%, and mean annual precipitation of 2000 mm. Water is least available between June and December (Diniz, 1986).

The vegetation of the study area was classified, following Veloso et al. (1991), as dense rainforest, mixed open rainforest with lianas and palms, and dense alluvial forest. The most deforested area is located in the northeastern portion of the municipality, in a logging zone where hundreds of sawmills until the end of the twentieth century. Following this peak in logging, land was converted to cattle ranching and cash crop farming, activities that now predominate the local landscape (Fearnside, 2001; Pezzuti and Silva, 2009).

2.2. Data collection

2.2.1. Biological sampling

The 50 stream reaches were sampled during the dry season (from June to August 2011) given that the intense precipitation typical of the rainy season impedes access to the study sites and the collection of many variables, including adult odonate specimens, given their ecophysiological demands (May, 1976, 1991; Corbet, 1999; De Marco et al., 2015). A 150 m reach of each study stream was delimited and then subdivided into 10 sections 15 m long, separated by 11 transects (following Peck et al., 2006). The physical characteristics of the habitat of the stream channel and the riparian vegetation were measured in each transect and along the sections. Each section was subdivided in three segments of 5 m for the collection of adult odonates, with only the first two segments of each section being sampled, resulting in a sample of 20 segments per stream. All the specimens collected at a stream were combined to form a single sample for that stream. The adults were collected using an entomological net (40 cm in diameter, with a 65 cm net and an aluminum handle 90 cm long). The mean sampling time at

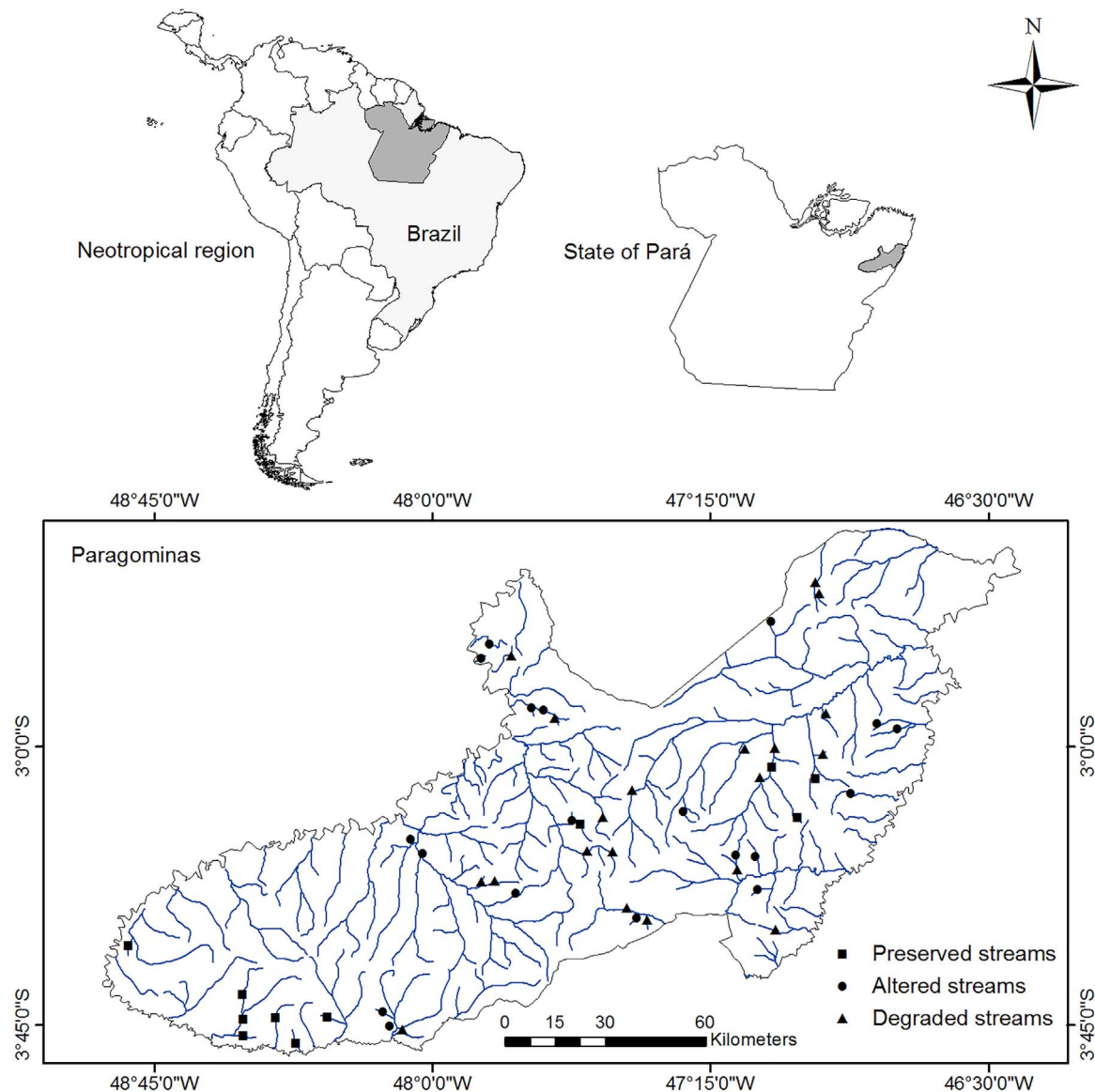


Fig. 1. Drainage network and distribution of streams according to the level of conservation (preserved, altered or degraded) sampled in the Eastern Amazon, Paragominas, PA, Brazil.

each site was one hour (following De Marco, 1998), an approach used successfully in previous studies (Juen and De Marco, 2011; Pinto et al., 2012; Oliveira Junior et al., 2015) for the rapid assessments of the biodiversity of streams.

Samples were collected between 10 a.m. and 2 p.m., and only when the stream was fully illuminated by sunlight. These conditions were necessary to ensure that all odonate groups (conformers, heliothermic and endothermic) were active during sampling (May, 1976, 1991; De Marco and Resende, 2002). The specimens were prepared in the field, using the protocol described by Lencioni (2006), being wrapped in paper envelopes and then washed in P.A. acetone (12 h for the zygopterans, and 48 h for the anisopterans). The acetone extracts fat and water from the specimens, which reduces the decomposition of the pigments. The specimens were subsequently dried under a heat lamp for 3 h. To keep the specimens dry, packets of silica gel were added to the storage containers.

The specimens were identified using taxonomic keys and species descriptions (Borror, 1945; Belle, 1988, 1996; Garrison, 1990; Carvalho and Calil, 2000; Costa et al., 2002; Lencioni, 2005, 2006; Garrison et al., 2006; Garrison et al., 2010), which were also compared directly with voucher specimens in the collection of the Museum of Zoology (Museu de Zoologia) of the Federal University of Pará (UFPA) in Belém,

Pará, Brazil. Whenever necessary, specimens were sent to the specialists Frederico Lencioni (Zygoptera) and Ângelo Pinto (Anisoptera) for identification. Once identified, the specimens were deposited as vouchers in collection of the UFPA Museum of Zoology in Belém.

A sample of at least 10 specimens in good condition were selected from those available for each species, representing the different environments surveyed, to estimate the mean dry weight of an individual. The specimens were dried in the open air, following Lencioni (2006), and then weighed using a precision balance with an accuracy of 0.005 g.

2.2.2. Physical integrity of the sites

The physical aspects of the environmental quality of each stream were evaluated using the Habitat Integrity Index (HII) of Nessimian et al. (2008). This protocol consists of 12 items that describe the environmental conditions, assessing characteristics that include patterns of land use adjacent to the riparian vegetation, the width of the riparian forest and its state of preservation, the condition of the riparian forest within a radius of 10 m, a description of the type of sediment and presence of retaining debris, the structure and erosion of the banks, the characteristics of the substrate of the stream bed, the aquatic vegetation and debris, and the distribution of rapids, pools, and meanders.

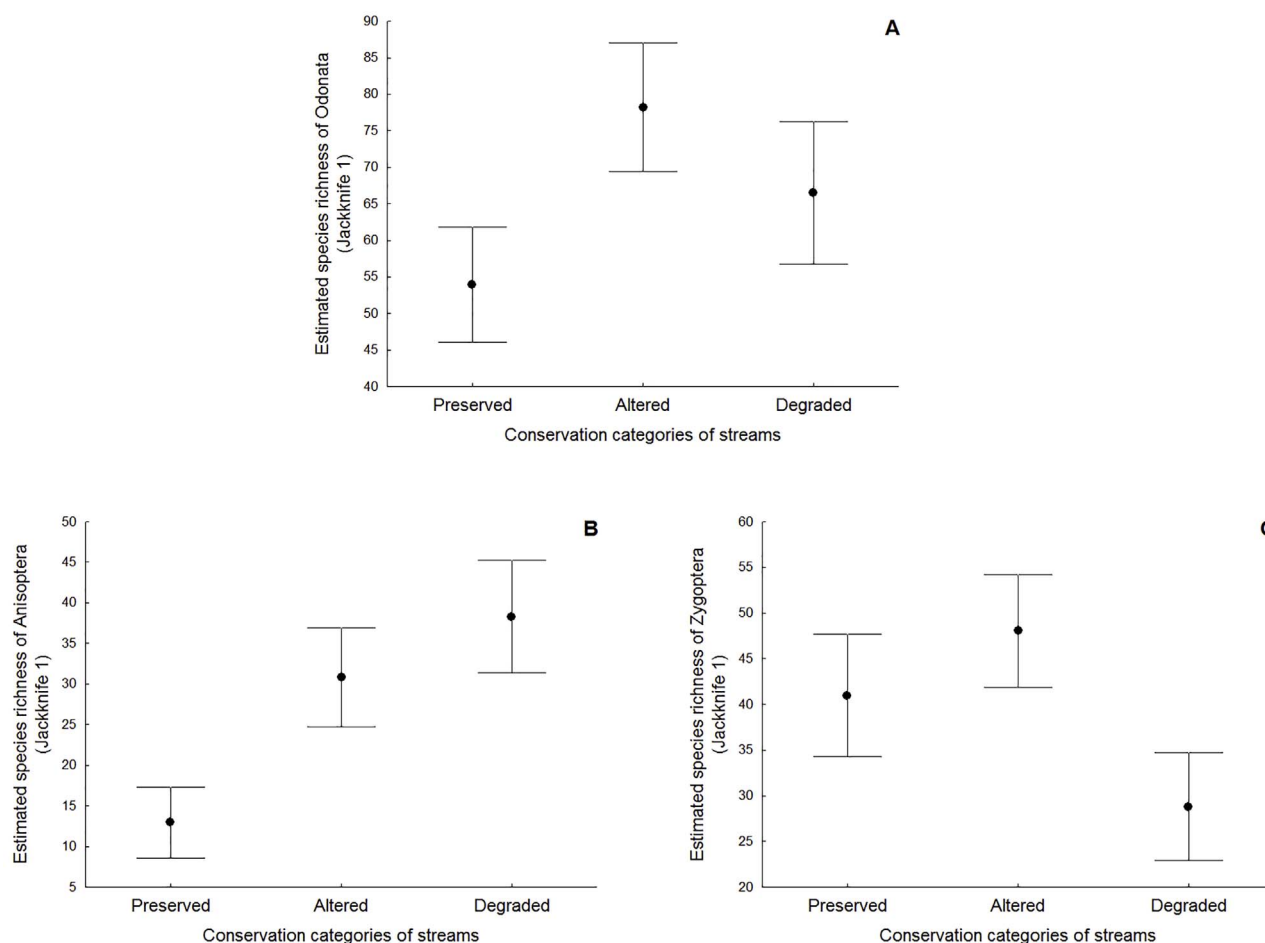


Fig. 2. Estimated species richness (according to the first-order jackknife estimator) (mean $\pm$ confidence interval) by conservation levels of the streams sampled in the eastern Amazon, Paragominas, PA, Brazil, considering (A) Odonata (B) suborder Anisoptera and (C) suborder Zygoptera.

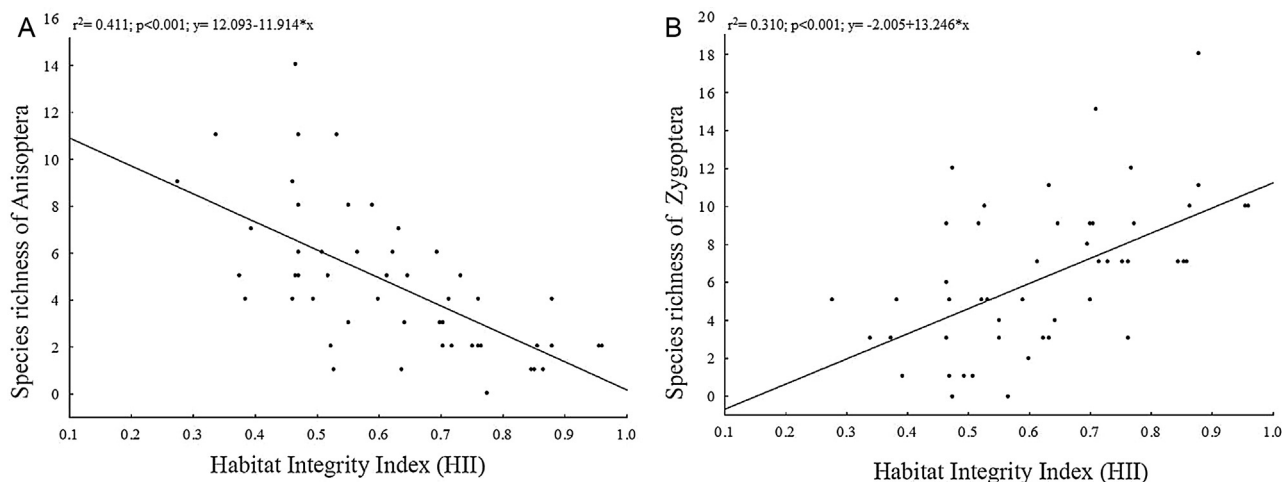


Fig. 3. Relationship between species richness of Odonata; A) Anisoptera; B) Zygoptera with Habitat Integrity Index (HII) of streams sampled in the Eastern Amazon, Paragominas, PA, Brazil.

Each of these items is scored on a scale of four to six points, and these values are summed and divided by the maximum possible score to provide a final value of 0–1 for the HII, with the highest values indicating the greatest stream integrity. These scores were divided into three classes – low, intermediate, and high integrity – based on field observations and the experience of the researchers. In the present study, the HII values ranged from 0.28 to 0.96, and the 50 streams were classified as degraded (low integrity) when the HII was less than 0.50

(N = 20 streams), altered (intermediate integrity) when the HII was 0.50–0.75 (N = 19), and preserved (high integrity) when the HII was above 0.75 (N = 11).

2.2.3. Riparian vegetation

The quantification of the structure of the riparian vegetation was based on Peck et al. (2006). Mean channel shading was estimated using a convex densiometer at the center of the channel, where four

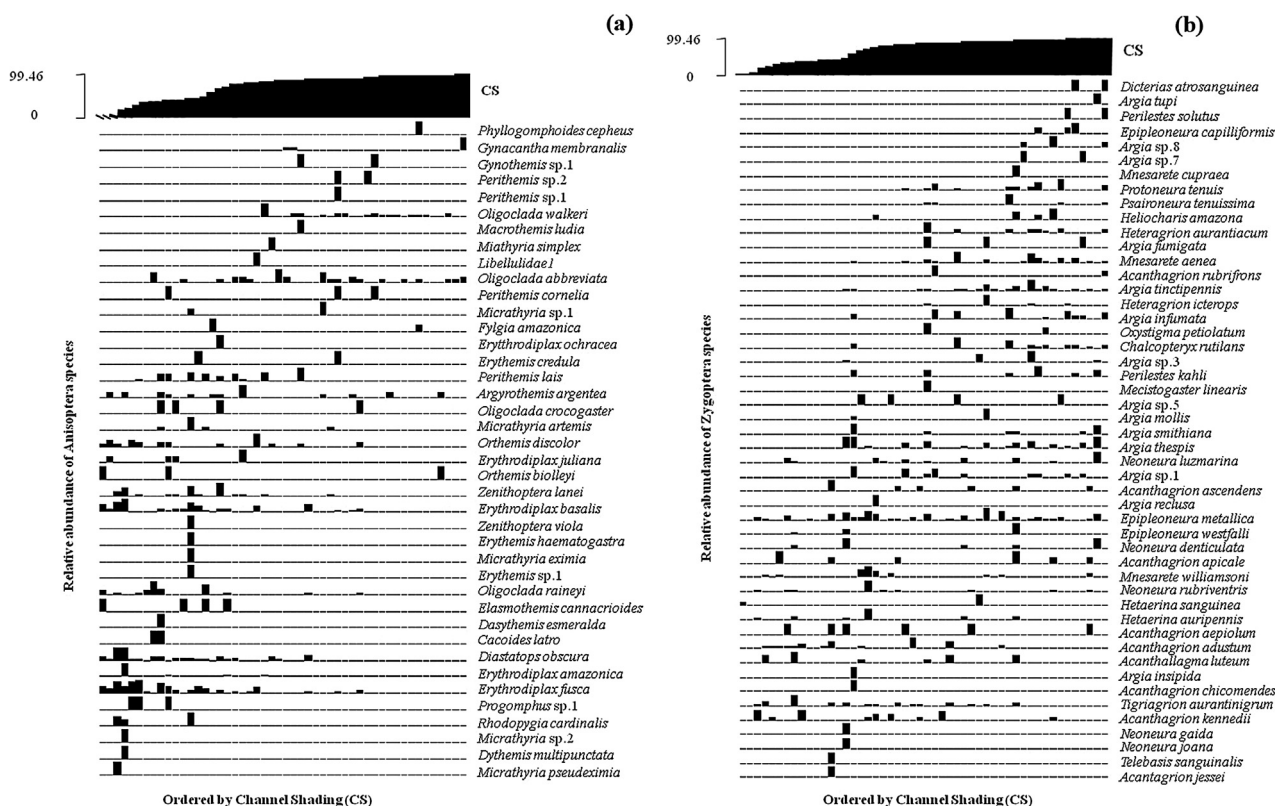


Fig. 4. Direct ordination of the relative abundance of the odonate species found in Paragominas stream sites, ranked by channel shading (CS). Species are ranked along the y axis by the mean reciprocal, weighted by the channel shading: (a) Anisoptera; (b) Zygoptera.

measurements were taken, facing upstream, downstream, and towards the left and right banks.

2.3. Data analysis

A first-order non-parametric jackknife estimator (Colwell and Coddington, 1994) was used to infer the total species richness of the streams, controlling for sampling effort. The sampling efficiency of the odonate species was evaluated using cumulative species curves (rarefaction curves) based on the data from the segments sampled in each section (Colwell et al., 2004).

The confidence interval of the species richness estimate (Gotelli and Colwell, 2001; Colwell et al., 2004) was used to evaluate the variation in species richness among sites based on their level of integrity (preserved, altered or degraded). A significant difference between groups was determined when the confidence interval of the means did not overlap between classes. A simple regression was calculated between odonate species richness and the HII value of each site to evaluate the response of the odonate community to the integrity of the physical habitat.

A multiple regression (Zar, 1999) was used to evaluate the response of species richness to changes in riparian vegetation cover. The relationship between the composition of odonate species and channel shading was assessed by ranking the relative abundance of the species (except those represented by fewer than three specimens or that were present at only one site) and compared this ranking with the percentage of shading at each site.

The approach of Warwick and Clarke (1995) was used to analyze the taxonomic diversity (Δ) and the taxonomic distinctness (Δ^*). This approach is based on the topological distance of phylogeny, in which the relationship between two species is estimated by the number of nodes or links (distance between branches) that separate them in a phylogenetic tree. The taxonomic diversity is calculated by the length

of the branches between two individuals of different species chosen at random from the sample. The taxonomic distinctness consists of modifying the taxonomic diversity to minimize the effects of dependence on the abundance of individuals, by weighting the length of the branches between any two individuals of different species chosen randomly from the sample.

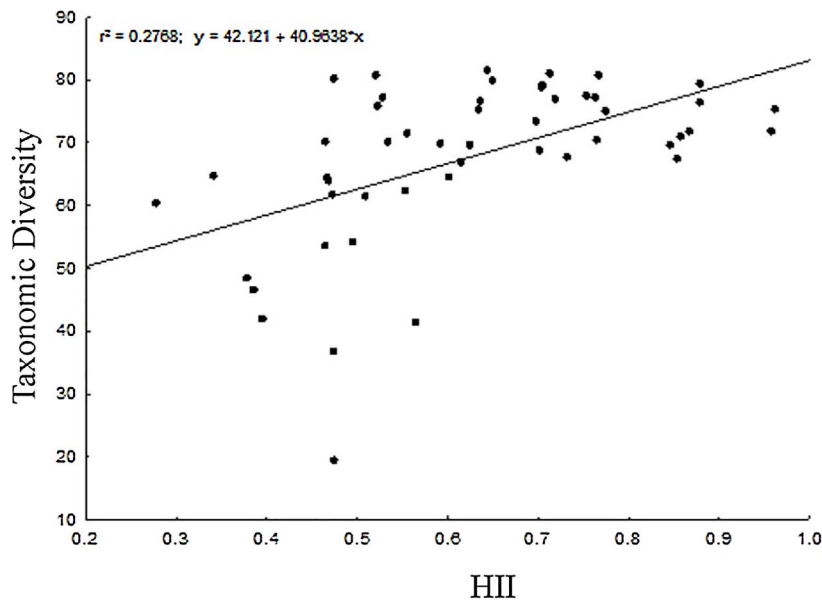
The phylogeny of the Odonata was compiled from a supertree created by Juen (2011), with information added from the new phylogenies published up to 2015, using the representation matrix based on the parsimony method (Baum and Ragan, 2004). Five levels of taxonomic resolution were used for this analysis – order, suborder, family, genus, and species, and individuals not identified to species were excluded from the analysis. Simple regression were used to evaluate the response of the taxonomic diversity (Δ) and distinctness (Δ^*) of the odonate assemblages to the variation in habitat integrity (HII).

The comparison between the abundance and biomass (ABC) curves at different levels of anthropogenic disturbance (preserved, degraded and altered streams) was based on the W statistic proposed by Warwick (1986). This analysis consists of a numerical summary of the ABC curve, with values ranging from -1 (abundance curve above biomass curve) to 1 (biomass curve above abundance curve).

The variation in the composition of the odonate species along the HII gradient was analyzed using a multivariate distance matrix regression, or MDMR (Zapala and Schork, 2006, 2012). This technique begins with the transformation of the raw data into a distance matrix. The Bray-Curtis dissimilarity index was the applied, the abundances were $\log(x + 1)$ transformed (Kruskal, 1964). The biological dissimilarity matrix was then compiled and modeled in relation to the continuous predictor (HII). A Pearson correlation matrix was compiled between the scores of the sample streams generated by the MDMR and the Habitat Integrity Index (HII) to determine the degree of dissimilarity in the species composition along the environmental gradient.

The Threshold Indicator Taxa Analysis – TITAN (Baker and King,

(A)



(B)

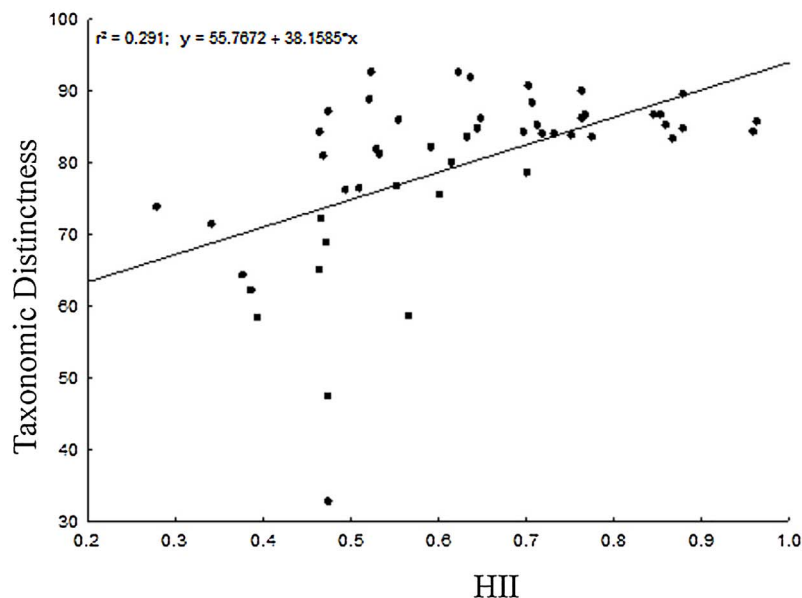


Fig. 5. Linear regression between the Odonata Taxonomic Diversity (A) and Taxonomic Distinction (B) values, and the Habitat Integrity Index (HII) values in streams sampled in the Eastern Amazon, Paragominas, PA, Brazil.

2010) was used to identify the species that responded to the disturbance gradient defined by the HII. This analysis aimed to detect variation in the distribution of the taxa along the continuous environmental gradient and to verify the existence of the thresholds at which the frequency and relative abundance of each taxon shift along this gradient (Baker and King, 2010). This analysis was based on the specificity and fidelity values of the Indicator species analysis, or IndVal (Dufrene and Legendre, 1997) and in the Change-Point Analysis, the nCPA (King and Richardson, 2003). The nCPA identified boundaries between

communities in relation to the variable tested, providing confidence intervals and identifying change points (positive or negative) for the taxa (Baker and King, 2010; King et al., 2011). The TITAN distinguishes negative (z-) from positive (z+) responses, reflecting the association between the taxa and the environmental gradient, and tracking the responses of the taxa in the community, which may either decline [sum (z-)] or increase [sum(z+)]. Each species would thus have either a low (z-) or high (z+) degree of association with the gradient (Baker and King, 2010). Bootstrapping was used to estimate the reliability and

Table 1Taxonomic Diversity (Δ) and Taxonomic Distinction (Δ^*) for the different conservation levels of the streams sampled in the Eastern Amazon, Paragominas, PA, Brazil.

Stream conservation level	Taxonomic Diversity (Δ)			Taxonomic Distinction (Δ^*)		
	Expected	Minimum	Maximum	Expected	Minimum	Maximum
All streams	83.645	19.553	81.684	79.135	32.838	92.751
Preserved Streams	79.852	61.572	74.678	72.346	73.674	85.750
Intermediate Streams	88.612	52.343	78.695	78.316	61.618	91.798
Degraded Streams	69.972	21.115	77.010	65.301	35.462	86.522

purity of the indicators, and the uncertainty associated with the change points (Baker and King, 2010).

A Principal Coordinates Analysis (PCoA) was used to sort the sites according to their species composition. This analysis was also based on the Bray Curtis index of the log transformed abundance values. Pearson correlation coefficients were then used to check for collinearity between the PCoA axis with the highest degree of explanation and the estimated species richness (Jackknife1), Taxonomic Diversity (Δ) and Taxonomic Distinctness (Δ^*).

The estimates of species richness were calculated in EstimateS (Statistical Estimation of Species Richness and Shared Species from Samples) version 7.5.0 (Colwell, 2005). All other analyses were run in the Vegan package of the R software (R Development Core Team, 2011). The script proposed by Baker and King (2010) was used to calculate the TITAN.

3. Results

3.1. Odonate community structure

A total of 1769 odonate specimens were sampled, representing nine families, 41 genera and 97 species. The suborder Zygoptera was represented by 961 individuals, distributed in six families (Calopterygidae, Coenagrionidae, Dicterididae, Megapodagrionidae, Perilestidae, and Polythoridae), 19 genera, and 56 species., while the Anisoptera contributed 808 individuals, distributed in three families (Aeshnidae, Gomphidae and Libellulidae), 22 genera, and 41 species.

The Coenagrionidae was the most abundant zygopteran family, with 650 specimens, and *Argia* was the most abundant zygopteran genus ($n = 230$ specimens), followed by *Acanthagrion* ($n = 72$) and *Tigriagrion* ($n = 26$). The Libellulidae was the most abundant anisopteran family, with 796 specimens, and *Erythrodiplax* ($n = 375$), *Oligoclada* ($n = 136$) and *Diastatops* ($n = 105$) were the most abundant genera.

3.2. Variation in odonate species richness according to stream integrity

The highest estimated odonate species richness was recorded in streams classified as altered (78.23 ± 4.45), with 12 more species, on average, than the degraded streams, and 24 more than the preserved ones (Fig. 2A). The species richness of the suborder Anisoptera was highest in the degraded streams (38.31 ± 3.51), with seven species more, on average, than the intermediate streams, and 25 species more than the preserved ones (Fig. 2B). By contrast, the estimated zygopteran species richness was highest in the altered streams (48.06 ± 3.13), with 19 species more, on average, than the degraded streams, and seven species more than the preserved ones (Fig. 2C).

The Habitat Integrity Index affected the species richness of both the Anisoptera ($r^2 = 0.411$; $p < 0.001$, Fig. 3A) and the Zygoptera ($r^2 = 0.310$; $p < 0.001$, Fig. 3B). As environmental integrity increased, the zygopteran species richness also increased, whereas anisopteran species richness decreased.

3.3. Relationship between riparian vegetation cover and the odonate assemblages

We found a significant relationship between the presence of riparian vegetation and the species richness of both the Anisoptera ($R^2 = 0.390$, $p < 0.001$) and the Zygoptera ($R^2 = 0.300$, $p = 0.006$). Reduced stream canopy cover generated an increase in anisopteran species richness (curve slope = -0.567 , $p < 0.001$), and a decrease in the number of zygopteran species (curve slope = 0.405 , $p = 0.006$). Species composition also varied systematically with the degree of channel shading, with a marked substitution of species occurring as the percentage of channel shading changed (Fig. 4). The relative abundance of anisopteran species was greater in less shaded streams (Fig. 4A), whereas the density of zygopteran species increased with increased shading (Fig. 4B).

3.4. Taxonomic diversity (Δ) and taxonomic distinctness (Δ^*)

The taxonomic diversity (Δ) and distinctness (Δ^*) varied considerably among the categories of habitat integrity, which indicates a shift in community structure along the anthropogenic disturbance gradient. While the Δ and Δ^* values were related systematically with the habitat integrity index (HII), the predictive value was intermediate ($r^2 = 0.526$; $p < 0.001$; $r^2 = 0.540$; $p < 0.001$; Fig. 5). Taxonomic diversity values ranged from 19.552 to 81.684, and those for distinctness, between 32.837 and 92.751 (Table 1). Altered streams had the highest taxonomic diversity and distinctness ($\Delta = 88.612$; $\Delta^* = 78.315$). The lowest taxonomic diversity ($\Delta = 69.971$) was recorded in the degraded streams, while the preserved streams had intermediate values ($\Delta = 79.851$; $\Delta^* = 72.346$).

3.5. Abundance and biomass of odonates according to stream integrity

The abundance curves were located above the biomass curves in the streams of all three integrity classes (degraded: $W = -0.114$; altered: $W = -0.047$; preserved: $W = 0.053$) (Fig. 6A–C). However, the difference was relatively discreet in the case of the preserved streams, as indicated by the W value. The same pattern was found in both the Anisoptera (degraded: $W = -0.294$; altered: $W = -0.230$; preserved: $W = -0.043$) and the Zygoptera (degraded: $W = -0.292$; altered: $W = -0.167$; preserved: $W = -0.137$), when analyzed separately.

3.6. Variation in species composition in relation to the environmental gradient

The species composition varied systematically along the environmental gradient established by the HII values (Table 2). Habitats of higher integrity have less similar communities, while those of lower integrity have much more similar communities ($r = -0.902$, $p < 0.001$).

The TITAN detected a marked shift among the taxa along the gradient established by the HII values, identifying 20 indicators from the 97 species analyzed. Twelve species (i.e. 60% of the 20 indicators identified) made up a group that was associated significantly with sites of high habitat integrity. The disturbance thresholds of these species

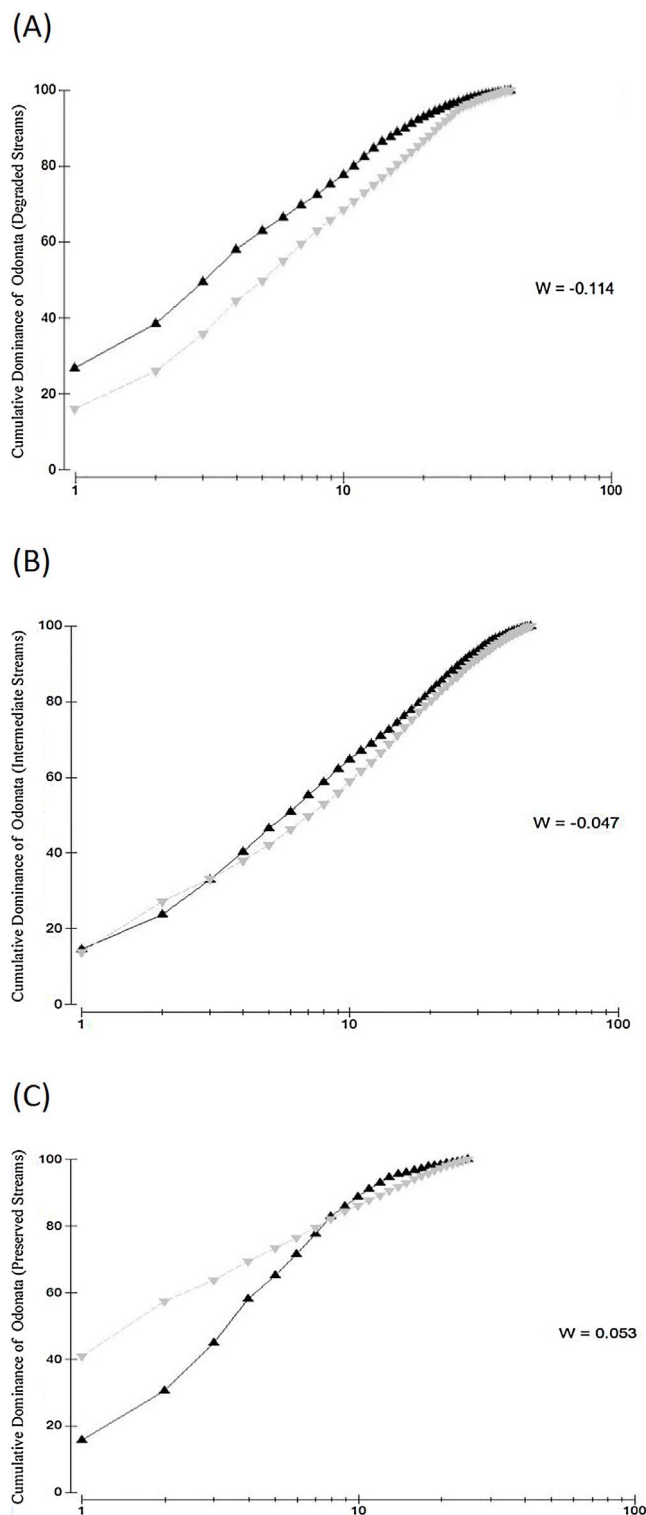


Fig. 6. ABC curves; Abundance (black triangle) and biomass (gray triangle) performed for Odonata order in: (A) degraded, (B) intermediate and (C) preserved streams. Sampled in Eastern Amazonia, Paragominas, PA, Brazil.

ranged from 0.47 to 0.85 (Fig. 7). Eight species (40% of the indicators) were associated with low habitat integrity, with thresholds ranging from 0.46 to 0.75.

3.7. Relationships among the parameters of the odonate assemblages

We did not find any significant relationship between the

Table 2

A Multivariate distance matrix regression (MDMR) was used to evaluate the association between the species composition of the Odonata community and the Habitat Integrity Index (HII) of stream sampled in the eastern Amazon region, Paragominas, PA, Brazil.

	DF	SS	MS	F	R ²	p
HII	1	3.684	3.684	12.668	0.208	< 0.001
Residuals	48	13.960	0.291	0.791		
Total	49	17.645	1			

composition of the assemblages (represented by the first axis of the PCoA) and estimated species richness ($r = -0.145$, $p = 0.315$). However, most of the variables were correlated significantly (Table 3). The taxonomic diversity and distinctness metrics were correlated positively ($r = 0.939$, $p < 0.001$). The composition of the assemblages was correlated negatively with taxonomic diversity ($r = -0.607$, $p < 0.001$) and distinctness ($r = -0.589$, $p < 0.001$).

4. Discussion

Adult odonates have highly specific habitat preferences. Given this, the most sensitive parameters for detecting the relationships between odonate assemblages and environmental integrity were the species composition, and taxonomic diversity and distinctness. These are probably the best measures because they incorporate information on the identity of the species at each site. Other variables, such as the estimated species richness and the abundance/biomass ratio represented by the ABC curve were less effective for the evaluation of the relationship between the odonate assemblages and anthropogenic impacts. In addition, as these measures provide no information of the characteristics of the species, they do not provide a reliable assessment of the conservation status of the site (Magurran, 2004).

Environmental integrity was an important element in the structuring of the two odonate suborders, with absolutely opposite effects. Zygopteran species were sensitive to changes in the physical integrity of these streams, while the anisopteran species appeared to have been favored by disturbance, given the increase in species richness observed in the more impacted streams. Given this, species richness per se did not provide a reliable measure of impact. One fundamental factor here may be the loss of the riparian forest (Voelz and McArthur, 2000), which has a direct effect on community structure and favors the Anisoptera over the Zygoptera (Oliveira Junior et al., 2015). Given this, the deconstruction of the odonate communities would be recommended, given that large groups often contain species with widely differing environmental requirements, which can hamper the detection of the effects of anthropogenic changes.

The increase in the incidence of sunlight and heat in degraded environments may account for the greater anisopteran species richness found in these environments (Juen et al., 2014). As these relatively large-bodied species are heliothermic, the greater incidence of sunlight favors their metabolism and their activity (Resende, 2010; Corbet and May, 2008; May, 1976). These findings indicate that the adults of many anisopteran species disperse to deforested areas, and avoid more shaded streams. The smaller-bodied zygopterans have higher conductance, i.e., the temperature of the body varies with that of the environment (Juen et al., 2014), and are thus associated with more pristine environments and with well-preserved riparian vegetation (Corbet, 1999; Rodrigues et al., 2016).

Zygopteran species also have narrower niches and are poor dispersers, whereas anisopterans are pioneers, and are excellent colonizers of open or temporary habitats created by disturbances (Clausnitzer et al., 2009), and these more generalist species typically migrate to disturbed sites. Given this, the number of anisopteran species may be the same or even higher in degraded and intermediate environments, masking the effects of habitat disturbance (Juen et al., 2014).

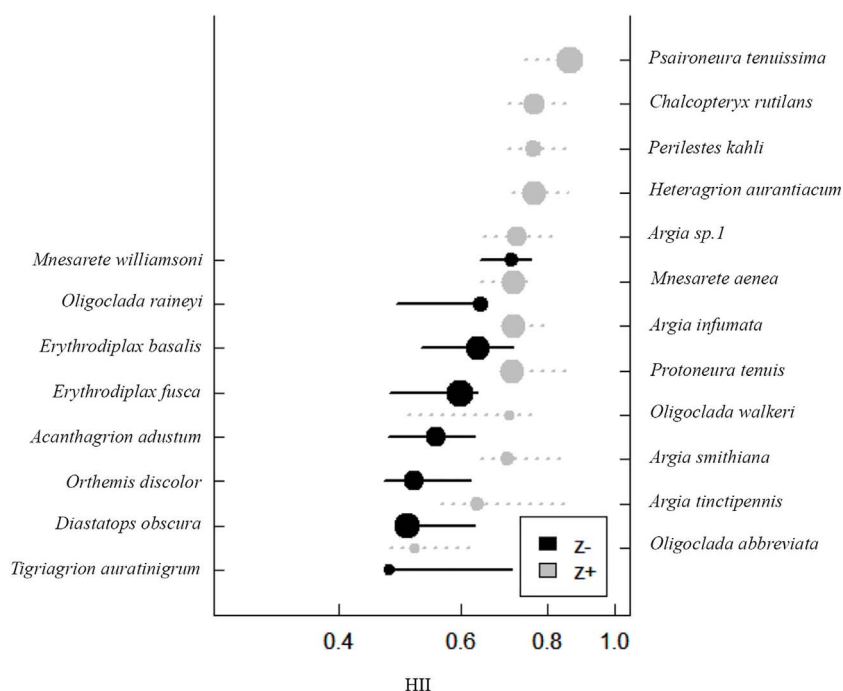


Fig. 7. Threshold Indicator Taxa Analysis (TITAN). Change points and 90% confidence interval of the significant indicator species ($n = 20$) along the disturbance gradient (HII). Significant indicator species are species with $\text{IndVal } p < 0.05$, purity > 0.95 and reliability > 0.95 for 500 bootstrap and 250 permutation replicates. Change points are represented by black and grey circles for species associated with low and high disturbance levels, respectively.

Table 3

Correlation between the metrics: First order Jackknife, Taxonomic Diversity (Δ), Taxonomic Distinction (Δ^*) and Species composition using the first axis of PCoA.

Odonata community metrics	β	EP	B	EP	t(48)	p
Jackknife $\times$ Diversity taxonomic (Δ)	0.448	0.129	0.958	0.276	3.475	0.001
PCoA I (composition) $\times$ Jackknife	-0.145	0.143	-0.759	0.748	-1.014	0.315
PCoA I (composition) $\times$ Taxonomic diversity (Δ)	-0.607	0.115	-0.247	0.047	-5.290	< 0.001
PCoA I (composition) $\times$ Taxonomic distinction (Δ^*)	-0.589	0.117	-1.594	0.315	-5.055	< 0.001
Jackknife $\times$ Taxonomic distinction (Δ^*)	0.307	0.137	0.594	0.266	2.232	0.030
Taxonomic diversity (Δ) $\times$ Taxonomic distinction (Δ^*)	0.939	0.050	1.035	0.055	18.863	< 0.001

In the degraded streams, the phylogenetic structure (taxonomic diversity and distinctness) of the assemblages was reduced in comparison with more preserved areas. Disturbed communities tend to have lower taxonomic distinctness because they are typically composed of more closely-related species than well preserved communities. Preserved communities tend to have taxonomically less related species, resulting in greater distinctness (Warwick, 1995). It is important to note that, like most diversity indices, taxonomic diversity and distinctness are relative measures for use in comparisons within a study, rather than absolute values for comparisons among studies (Warwick and Clarke, 1995). Taxonomic distinctness provides a highly sensitive univariate measure of changes in community structure. These metrics have been widely used in studies of marine ecology for the assessment of community integrity and the conservation status of habitats (Warwick and Clarke, 1995; Hall and Greenstreet, 1998; Rogers et al., 1999; Brown et al., 2002; Leonard et al., 2006).

A number of studies have shown that comparative abundance/biomass curves can be effective for the assessment of environmental impacts in aquatic systems (Carvalho et al., 2013). This approach was not effective in the present study of odonates, however. Similar results were obtained for both the order (Odonata) as a whole, and the two sub-orders (Anisoptera and Zygoptera), with the abundance curves predominating over the biomass curves. This pattern is theoretically consistent with the conditions found in degraded areas (Warwick and Clarke, 1995). One possible explanation for the observed pattern in preserved areas is the temporary presence of species that visit these areas only to forage, and are not resident. An alternative explanation is a temporary increase in the abundance of individuals from neighboring, impacted areas. Species found in these impacted areas are typically

endothermic or heliothermic, and are favored by the more open canopy resulting from the removal of the riparian vegetation, which enables them to heat their bodies more efficiently (Calvão et al., 2013). Once active, they are able to disperse to more preserved areas, where resources may be more abundant, to forage, contributing to the observed pattern in community structure.

By contrast, the composition of the odonate assemblages provided a reliable measure of anthropogenic impact, given that the composition of these assemblages varied systematically along the integrity gradient. Rather than simply changing the number of species, then, this dynamic alters interspecific relationships, including habitat selection, leading to the loss of certain species, and the addition of others (Monteiro-Junior et al., 2013). Colonization by new species, which are usually generalists and good competitors, tends to result in the elimination of resident species (De Marco and Resende, 2004). This pattern of species turnover would be expected, given the thermoregulatory characteristics of tropical species (De Marco et al., 2005). Large-bodied anisopterans, such as *Orthemis discolor* Burmeister, 1839 (Monteiro-Junior et al., 2014) and species of the *Eritrodiplax* (Carvalho et al., 2013), are expected to occur in open, sunny areas due to their high thermoregulation efficiency. The only zygopteran species associated with degraded environments was *Acanthagrion adustum* Williamson, 1916, a species found in lentic environments, such as swamps and pools, with larvae associated with macrophytes and debris (Carvalho, 1991; Carvalho and Nessimian, 1998; De Marco, 1998). This species is related to degraded environments because of the increase in the amount of macrophytes resulting from the impoundment of most degraded streams, associated with activities such as roadbuilding (Ribera and Vogler, 2000; Von Sperling and de Lemos Chernicharo, 2005).

By contrast, most zygopteran species are restricted to areas with extensive canopy cover and shading, due to their smaller body size and associated ecophysiological restrictions, making them more sensitive to variations in the environment, and their greater habitat specialization (Corbet, 1999; Corbet and May, 2008). The females of the genus *Chalcopteryx*, for example, require specific habitats for oviposition and are thus only found in preserved environments (Oliveira Junior et al., 2015). The zygopteran species collected in the present study, such as those of the genus *Argia*, were also associated with well-preserved environments, which have an abundance of submerged plants and riparian vegetation, as observed by Monteiro-Junior et al. (2014). These species typically use the vegetation, and tend to be eliminated when the environment is disturbed, and the availability of perches and potential refuges decreases.

5. Conclusion

This study evaluated the effectiveness of the different measures and analytical tools available for the assessment of environmental quality, using odonates as bioindicators. The species richness of adult odonates did not provide an accurate assessment of the impact of habitat degradation on the community, given that the highest species richness was recorded in habitats with an intermediate level of degradation. A similar pattern was observed in the suborder Zygoptera, while the greatest species richness of the suborder Anisoptera was recorded in degraded habitats. By contrast, the variation in species composition, and taxonomic diversity and distinctness was associated systematically with the conservation status of the study streams. As expected, the species composition recorded in intermediate and degraded streams was distinct from that found in the preserved streams. From this perspective, assemblages of adult odonates provide effective indicators of the ecological conditions found in logic Amazonian environments. Given the conflicting responses of the species of the two suborders to habitat disturbance, measures of species composition appear to be the most effective parameters for the assessment of the impact of anthropogenic disturbance on odonate assemblages.

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